

CHAPTER 4 Assignment

3 The CPU checks for interrupts at the beginning of every -fetch-decode-execute cycle. It suspends current process when an interrupt signal is detected. The interrupt request number is used to index the interrupt vector table to get the address of the interrupt service routine which is placed into the program counter. Execution resumes within the ISR and upon completion the CPU restores the saved information from the interrupted program.

5 a) MM is byte addressable.

$$4M \times 16$$

$$4M \text{ with 2 words} = 8M \text{ bytes.}$$

$$= 2^3 \times 2^{20}$$

$$= 2^{23} \text{ bytes.}$$

$$= 23 - \text{bits needed}$$

b) MM is word addressable

$$4M \times 16$$

$$= 2^2 \times 2^{20}$$

$$= 2^{22}$$

$$= 22 - \text{bits needed}$$

6 a) MM is byte addressable?

$$1M \times 8$$

$$= 1 \times 2^{20} \text{ bytes}$$

$$= 2^{20} \times 2^0$$

$$= 20 \text{ bits}$$

b) MM is word addressable

$$1M = 2^{20} \text{ bytes.}$$

$$= 20 - \text{bits}$$

11 a) 16×20

$$= \frac{16 \times 2^{20} \times 16}{512 \times 2^{10} \times 8}$$

$$= \frac{2^{24} \times 2}{2^{19}}$$

$$= 2^5 \times 2$$

$$= 64 \text{ RAM chips.}$$

b) $= \frac{16 \text{ bits}}{8 \text{ bits}}$

$$= 2 \text{ banks}$$

c) Address bits

$$\log_2(512) + \log_2(2^{10})$$

$$= 9 + 10$$

$$= 19 \text{ bits.}$$

d) 2 banks:

e) $19 + \log_2(2) = 19 + 1$

$$= 20 \text{ bits for addressing}$$

f) 14 (E in hex) would not be in the specific memory

g) 14 (E in hex) for low-order interleaving would not be in a specific memory location.

$$\begin{aligned}
 \text{12a) No of RAM chips} &= \frac{32 \text{ G} \times 64}{1 \text{ G} \times 16} \\
 &= \frac{32 \times 64}{16} \\
 &= 128 \text{ RAM chips.}
 \end{aligned}$$

b)

$$\begin{aligned}
 \text{b) No of banks} &= \frac{128}{4} \\
 &= 32 \text{ banks}
 \end{aligned}$$

$$\begin{aligned}
 \text{c) No of lines} &= \frac{\text{No. of data lines}}{\text{No. of chips per bank.}}
 \end{aligned}$$

$$= \frac{16}{4}$$

$$= 4 \text{ data lines.}$$

$$\text{d) Total No of words} = \frac{\text{Total Mem size}}{\text{Word size}}$$

$$= \frac{32 \text{ G}}{64}$$

$$= \frac{32 \times 2^{30}}{64}$$

$$= 2^{29}$$

$$= 29 \text{ words}$$

$$13) a) 2^n \geq 150$$

$$n \geq \log_2(150)$$

$$n \geq \frac{\log(150)}{\log(2)}$$

$$= 7.215$$

$$\approx 8$$

= 8 bits for the opcode

$$b) 24 - 8$$

= 16 bits for the instruction.

$$c) = 2^{16}$$

= 65 536 words.

$$d) 2^{24} - 1$$

$$= 16\,777\,216 - 1$$

$$= 16\,777\,215$$

~~24~~

$$16\,777\,216 \text{ M}$$

$$= 2^8 \times 2^{20}$$

= 268 435 456 bytes.

$$b) \text{ byte addressable} = 2^{28} \times 2^2$$

$$= 2^{30}$$

= 30 bits.

$$c) \text{ word addressable} = 2^{28}$$

= 28 bits.

25) a) 00010010 000101

- 0011 0010 0000 1010

- 01 00 0010 0000 1011

- 0010 0010 0000 1100

- 1000 10 00 0000 0000

1001 0010 0000 1000

1001 0010 0000 1100

0111 0000 0000 0000

b) Symbol	Memory Address
Done	208
Base	209
Offset	20A
One	20B
Addr	20C

c) 0208

27) a) Halt

b) Output 001100110000

c) Subt 111101001111

28) a) clear 010111000000

b) Load 101110010011

c) N/A